

The Resting Potential

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

The resting potential describes the stable voltage difference between the intracellular and extracellular space of an unstimulated neuron. Typically, the membrane potential is about -70 mV. It is based on an asymmetric ion distribution, the selective permeability of the membrane and the continuous contribution of the sodium-potassium pump. At rest, the cell membrane is not equally permeable to all ions. Mainly potassium leak channels (and a few sodium leak channels) are open, while voltage-gated potassium and sodium channels remain closed. As a result, the membrane is mainly permeable to potassium ions. Potassium diffuses outward along its concentration gradient, while relatively more negatively charged particles remain inside the cell. This builds up an electrical counterforce. The resting potential therefore results from the balance between the chemical concentration gradient and electrical attraction (= chemiosmotic and electrostatic forces).

The sodium-potassium pump helps maintain the ion gradients by transporting three sodium ions out of the cell and two potassium ions into the cell while consuming ATP. The resting potential is therefore a dynamic, energy-dependent stabilised state. Without it, excitability, action potentials and directed information conduction along the neuron would not be possible.

Task 1: Charge distribution and selective permeability

Describe the charge distribution between the inside and outside of the cell and explain the importance of the selectively permeable membrane for the formation of the resting potential.

Task 2: Importance of the sodium-potassium pump

Justify why the sodium-potassium pump is important for the long-term maintenance of the resting potential.

Task 3: Hypothesis about the effect of poison

Create a hypothesis about the effect potassium cyanide poisoning would have on the resting potential.
